

1. What is Mobile Development?

it is the process of creating applications for smartphones and tablets

2. What is Native Development?

it means building an app for one system only like Android or iOS using its own language

3. What is Cross Platform?

it is building one app using one codebase that works on multiple platforms like Android and iOS

4. Mention three Cross Platform frameworks.

1- flutter 2- React Native 3- Xamarin

5. What are the native languages for Android and iOS?

Android: Kotlin iOS: Swift

6. Explain one advantage of Flutter.

it uses a single codebase to build apps for both Android and iOS

7. Explain one difference between Flutter and React Native.

Flutter uses Dart programming language
React Native uses JavaScript programming language

Write the correct Dart data type for each value.

1.

5

Answer: int

3.

`"Kareem"`Answer: string

5.

`["Ahmed", "Ali"]`Answer: List string

2.

3.14

Answer: double

4.

`true`Answer: bool

6.

```
{  
  "name": "Ahmed"  
}
```

Answer: Map string,

Write Dart code for each of the following.

1. Create an integer variable called age.

Write your Dart code here

```
int age = 20;
```

2. Create a String variable called studentName.

Write your Dart code here

```
String studentName = "fady";
```

3. Create a bool variable called isStudent.

Write your Dart code here

```
bool isStudent = true;
```

4. Create a double variable called salary.

Write your Dart code here

```
double salary = 1500.50;
```

5. Create a List of three names.

Write your Dart code here

```
List<String> Names [ "Ahmed", "Ali", "omar" ];
```

6. Create a Map containing: name, age, city

Write your Dart code here

```
Map<String, dynamic> Person = {  
  "name": "fady",  
  "age": 20,  
  "city": "cairo"  
};
```

1. What is the difference between var and dynamic?

Var: Type is set once and cannot change

dynamic: Type can change at any time

2. When should you use dynamic?

receiving unknown data, like JSON from an API

3. Write one example using var.

Write your Dart code here

```
var Name = "Ali" // Automatically set as string
```

4. Write one example using dynamic.

Write your Dart code here

```
dynamic Value = 10;
```

```
Value = "Halo"; // type changed from int to string
```

1. What is the difference between final and const?

final: set once at runtime

const: set at compile time

2. When is final evaluated?

at runtime

3. When is const evaluated?

at compile time

4. Write one example using final.

Write your Dart code here

```
final String name = "Ali";
```

5. Write one example using const.

Write your Dart code here

```
const double pi = 3.14;
```

Write Dart code only for each instruction below.

1. Create a List of five cities.

Write your Dart code here

```
List<String> cities = ['Cairo', 'Alex', 'Giza', 'Luxor'];
```

2. Add one city.

Write your Dart code here

```
cities.add('suez');
```

3. Remove the third city.

Write your Dart code here

```
cities.removeAt(2);
```

4. Insert a city at index 2.

Write your Dart code here

```
cities.insert(2, 'Tanta');
```

5. Print the first item.

Write your Dart code here

```
print(cities.first);
```

6. Print the last item.

Write your Dart code here

```
print(cities.last);
```

7. Print the length of the List.

Write your Dart code here

```
print(cities.length);
```

8. Check if the List contains "Cairo".

Write your Dart code here

```
print(cities.contains('Cairo'));
```

9. Sort the List.

Write your Dart code here

```
cities.sort();
```

10. Clear the List.

Write your Dart code here

```
cities.clear();
```

7 List Functions

10 Marks

Write Dart code using each of the following functions.

1. Use map()

Write your Dart code here

```
List<int> Numbers = [1, 2, 3, 4, 5];
```

2. Use where()

Write your Dart code here

```
var doubled = Numbers.map((n) => n * 2);
```

3. Use any()

Write your Dart code here

```
bool hasEven = Numbers.any((n) => n % 2 == 0);
```

4. Use every()

Write your Dart code here

```
bool allPositive = Numbers.every((n) => n > 0);
```

5. Use forEach()

Write your Dart code here

```
Numbers.forEach((n) => print(n));
```

1. Create a Student Map containing: name, age, grade

Write your Dart code here

```
Map<String, dynamic> student = {
  "Name": "fady",
  "age": 20,
  "grade": "A"
};
```

2. Print the student's name.

Write your Dart code here

```
Print(student["Name"]);
// Print(student["Name"]);
```

3. Update the age.

Write your Dart code here

```
student["age"] = 23;
```

4. Add a new key called address.

Write your Dart code here

```
student["address"] = "Cairo";
```

5. Remove grade.

Write your Dart code here

```
student.remove("grade");
```

1. What is Compile Time?

the time before the program runs (when code is checked for errors)

2. What is Runtime?

the time when the program is actually running

Without looking at your notes, write a complete Dart program that demonstrates ALL of the following:

✓ Variables

✓ bool

✓ Map

✓ final

✓ Printing output

✓ int

✓ String

✓ var

✓ const

✓ double

✓ List

✓ dynamic

✓ List operations

Write your code below.

Write your complete Dart program here

```
void main() {
  int age = 20;
  double score = 95.5;
  String name = "Ahmed";
  bool isPassed = true;
  var city = "Cairo";
  dynamic data = 100;
  final String country = "Egypt";
  const double = 3.14;
  List<int> nums = [1, 2];
  nums.add(3); // List operation
  Map<String, int> marks = {"Math": 100};
  print(name);
}
```

B Challenge Question (Bonus)

Bonus Marks

Create a Student Management program that:

- Stores three students inside a List.
- Each student is represented as a Map.
- Print every student's name.
- Add a new student.
- Remove one student.
- Print the final List.

(Write the complete code.)

Write your complete Dart program here

```
void main() {  
  // 1. List of Maps for 3 students  
  List<Map<String, String> students  
    { "name": "Ali" }  
    { "name": "Omar" }  
    { "name": "Sara" }  
  };  
  // 2. Print every student's name for  
  (var s in students) {  
    print(s["name"]); }  
  // 3. Add a new student  
  students.add({ "name": "Mona" });  
  students.removeAt(0);  
  print(students);  
}
```